Department of Computer Science and Information Technology

Name				Id	
Course	CSCI430L Operating Systems Lab				
Date		**Time**		**Duration**	
Pages		**Calculators**	Not Allowed	**Documents**	Not Allowed

Question 1 [50 Points]

1. Write a class named **GenerateArrayThread** that generate an array of N integers randomly by values less than or equals to 100.
2. Write a class named **ReverseArrayThread** that reverse the elements of an array of N integers.
3. Write a class named **EqualsArrayThread** that test if 2 integer arrays are equals. That means they have the same length and the same elements.

Write a main that:

a. Prompts the user to enter the length N of the array.
b. Generates an array of N elements.
c. Find the reverse of the generated array in part b).
d. Finally, the main should test if the generated array (a) is palindrome. That mean it is equals its reverse

Question 2 [50 Points]

You can use Cramer's rule to solve the 2 ✕2 following system of linear equation:

$$\begin{aligned} ax + by &= e \\ cx + dy &= f \end{aligned} \quad x = \frac{ed - bf}{ad - bc} \quad y = \frac{af - ec}{ad - bc}$$

a. Write a Thread **EvaluateTerm** which evaluate an expression of form ab-cd. This Thread should have four data fields a,b,c and d.
b. Write a Thread called **EvaluateFraction** which evaluate an expression of from a/b. This thread should have 2 data fields a and b. This thread **must use** 2 instances of the thread **EvaluateTerm**.
c. Finally, the main class prompt the user to enter 6 reals numbers a, b, c, d, e, and f and displays the result. If ad - bc is 0, report that "**The equation has no solution**". You **must use** the threads of parts a and b.

```
Enter a, b, c, d, e, f: 9.0 4.0 3.0 -5.0 -6.0 -21.0 Enter
x is -2.0 and y is 3.0
```

```
Enter a, b, c, d, e, f: 1.0 2.0 2.0 4.0 4.0 5.0 Enter
The equation has no solution
```